

AYMANE AIT DADS

AI Software Developer Intern | Python · REST APIs · Cloud-Native CI/CD · Machine Learning

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PROFESSIONAL SUMMARY

Data Science engineering student at EURECOM (Master-level) who reduced manual network analysis time by 60% at Orange Maroc by building automated ML pipelines on 100,000+ fiber-optic records and delivering actionable insights to a network optimization team via Power BI dashboards. Ranked 17th out of 3,000+ teams in the ongoing NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by engineering a full LLM fine-tuning pipeline in Python with Hugging Face Transformers, demonstrating the technical depth and initiative SAP's AI software roles demand. Brings strong proficiency in Python, RESTful API design, Git-based source control, and cloud-native tooling - including Docker and REST frameworks - developed through academic projects spanning computer vision, NLP, and distributed ML systems. Aims to contribute production-quality AI features within SAP SuccessFactors' agile engineering teams while deepening expertise in scalable, cloud-compliant distributed systems across IaaS platforms.

CORE COMPETENCIES

RESTful API Design | Cloud-Native Deployment | Python Proficiency | Docker & Kubernetes | Agile Development | CI/CD Pipelines | Distributed Systems | Source Control (Git)

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built automated Random Forest and K-Means **ML pipelines on 100,000+ fiber-optic network records** (upload speed, latency, jitter), classifying network quality across thousands of nodes and eliminating manual bottlenecks.
- Engineered a clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for infrastructure planning.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation workflows built entirely in Python with Scikit-learn and Pandas.
- Delivered stakeholder presentations translating model outputs into maintenance recommendations, visualized via **Power BI** dashboards for non-technical business audiences.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in an ongoing NVIDIA reasoning challenge by fine-tuning a 30B-parameter Mamba-Transformer with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) using Python and Hugging Face Transformers.
- Ran **controlled ablation experiments** across learning-rate schedules, packing strategies, and synthetic augmentation - and built custom cipher parsers and 8-bit operation solvers - revealing that reasoning-style consistency outweighs data volume in LLM supervised fine-tuning.

Python · PyTorch · Hugging Face Transformers · Unsloth · vLLM · Git

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · 1st in Class Leaderboard

- Ranked **1st in EURECOM class** on a private Kaggle leaderboard (0.791 AUC) by fine-tuning a CLIP ViT-L/14 (428M params) with a custom MLP head, dual learning rates, and 10-view test-time augmentation across 250,000 training images.
- Submitted independently to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; identified that 1-epoch training (0.793 AUC) outperformed 4-epoch training (0.767 AUC), demonstrating superior out-of-distribution generalization.

Python · PyTorch · OpenCLIP · FP16 Mixed Precision · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Cloud Computing, Statistics, Web Applications, Image Security

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

Programming & APIs: Python (expert) | SQL | Bash | RESTful API Design | Git | Docker (basic) | Linux

ML / AI & Data Science: PyTorch | Scikit-learn | Hugging Face Transformers | Pandas | NumPy | Feature Engineering | Ensemble Methods

Cloud, MLOps & Visualization: FP16/BF16 Mixed Precision | PostgreSQL | Power BI | Matplotlib | Seaborn | Kaggle GPU | Ablation Studies